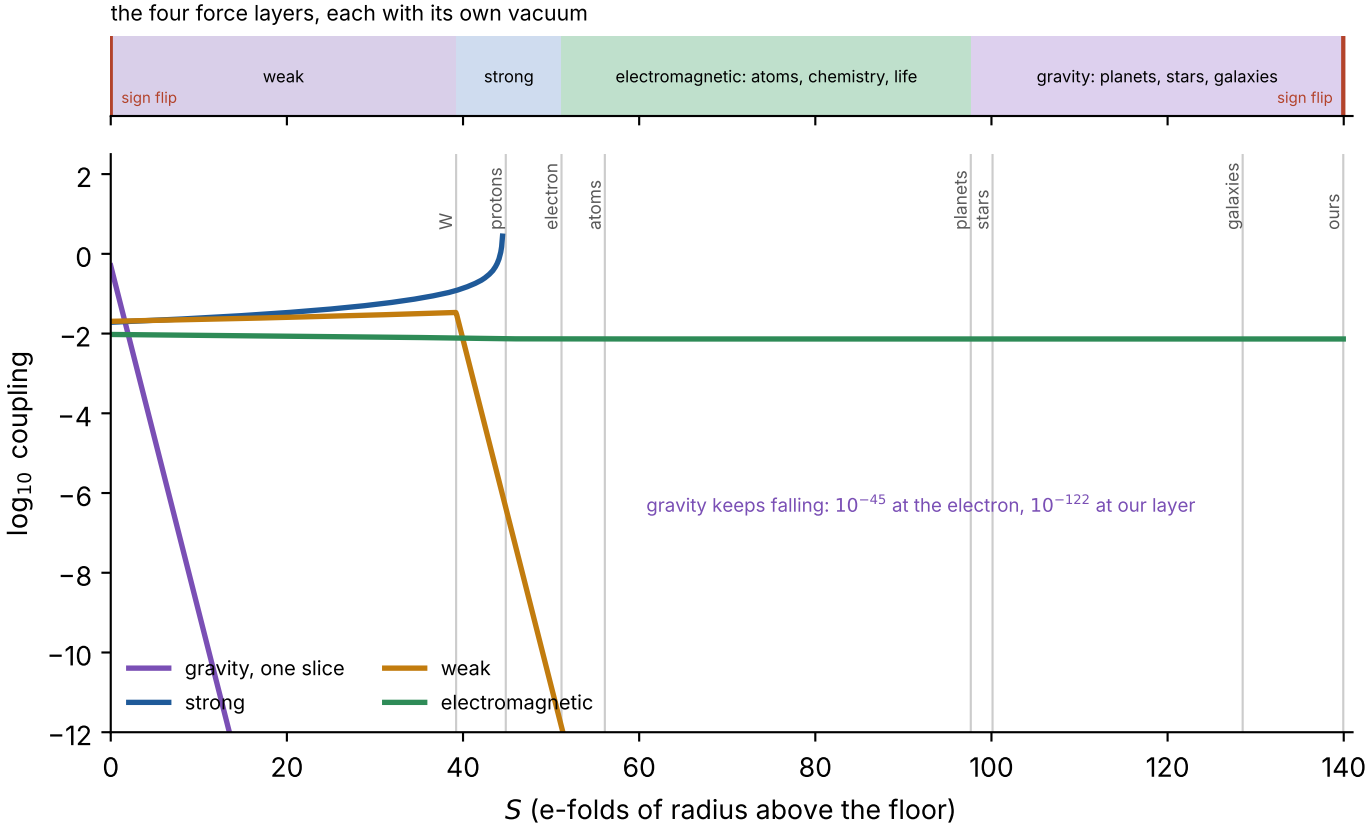
the four force layers, each with its own vacuum
weak
strong
electromagnetic: atoms, chemistry, life
gravity: planets, stars, galaxies
sign flip
sign flip
W
protons
electron
atoms
planets
stars
galaxies
ours
gravity keeps falling: 10^{-45} at the electron, 10^{-122} at our layer
gravity, one slice
strong
weak
electromagnetic
$\log_{10}$ coupling
2
0
−2
−4
−6
−8
−10
−12
0
20
40
60
80
100
120
140
S (e-folds of radius above the floor)